Embed PowerFX in your .NET Apps!

Intro

- Juan Pablo Tarquino
- Software Developer with about 20 years of experience
  - LinkedIn
    - https://www.linkedin.com/in/juantarquino
  - Blog (not updated often ☹)
    - https://jptarqu.blogspot.com
- Sample repo and slides for this session:
  - https://tinyurl.com/PowerFX-NET

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Today’s Talk

- Problem scenario
- Intro to PowerFX
- Why embedding PowerFX is a solution to the problem
- How to run the PowerFX interpreter from .NET
- How to extend the PowerFX interpreter

Once upon a time in Pizzas’R’Us™…

- Pizzas’R’Us™ is a small but profitable company, trying to be as nimble as possible, providing pizza services to restaurants and super-markets.

Problem scenario

- Pizzas’r’us
  - Small, fast evolving company
  - Small development team
- Alexandria – Lead Dev
  - Many changes requested from Business Units
  - Business Units trying to adapt to evolving business environment
  - The biggest part of the changes are business rule changes
  - Alexandria feels like the bottleneck

Business Logic vs Other logic

- Business logic is the logic that is unique to the business problem you are trying to solve.
- It can be very unique
  - Examples
    - Logic to drop a message in a queue is common across industries
    - Logic that specifies that customers with certain number of purchases get a discount is very specific.
- It can be executed in a vacuum with no direct dependencies on external resources

Solution Research

- Most requests are for Business rules changes
- Business users familiar with Excel
- A larger IT team that are not Pro devs, but work in PowerPlatform
  - PowerApps
  - PowerAutomate

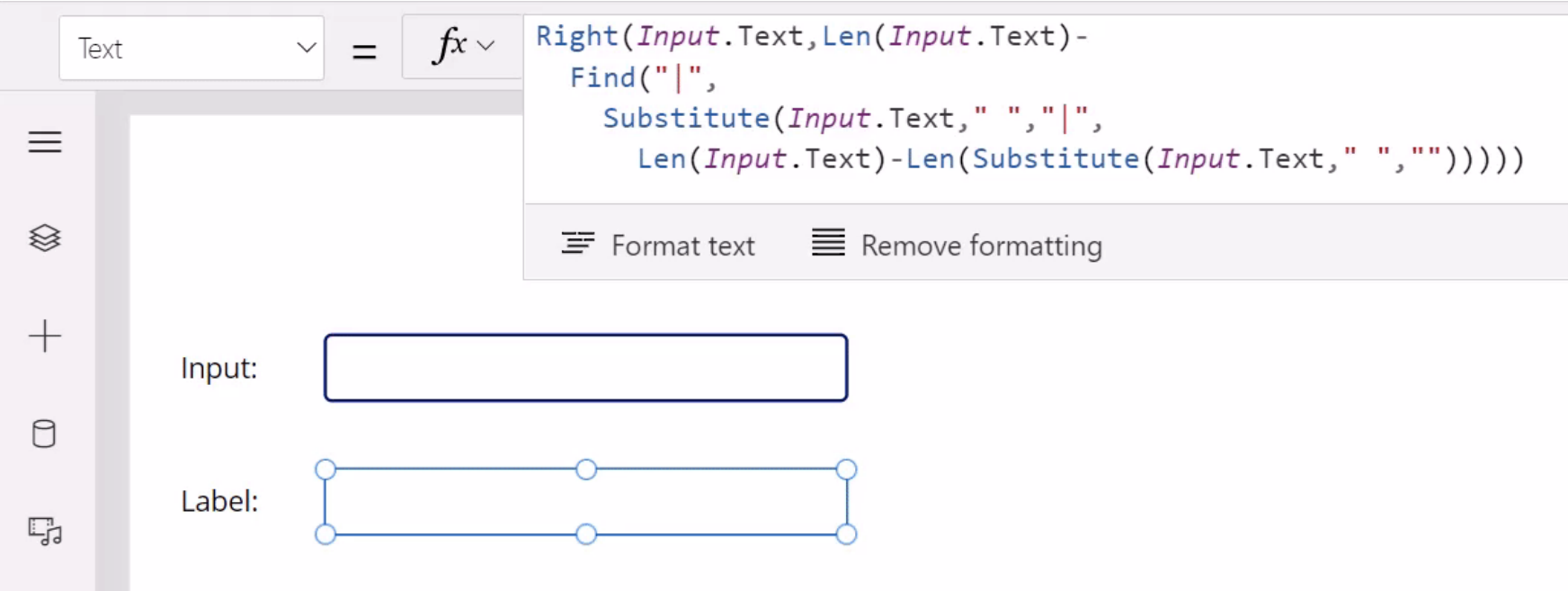
Solution Space

- Business rules engine frameworks
  - Most are Java based, not a good fit for team
  - Another paradigm for business users to learn
- Microsoft's Rules Engine
  - https://github.com/microsoft/RulesEngine
  - Unfamiliar syntax for existing org
- Graphical DSL
  - Classic Windows Workflows
- PowerFX

Intro to PowerFX

- Business user friendly
- Open sourced by Microsoft
- Similar to Excel formulas
- Use in many parts of the Power Platform
  - Many Experts, Tutorials, Articles, etc
- https://learn.microsoft.com/en-us/power-platform/power-fx/overview

PowerFx


Text
=
fx
Right(Input.Text,Len(Input.Text)-
Find("|",
Substitute(Input.Text," ","|",
Len(Input.Text)-Len(Substitute(Input.Text," "))))))
Format text
Remove formatting
Input:
Label:

Why we should consider using PowerFX from our .NET apps

- Tap into the power of PowerFX
  - Business users already familiar
  - Answers on how-to already on Internet
  - Co-pilot support
- Execution at runtime add dynamism to your app
- Business/Mapping rules

How to run the PowerFX interpreter from .NET

- Interpreter from Microsoft
  - https://github.com/microsoft/Power-Fx
- Microsoft Samples:
  - https://github.com/microsoft/power-fx-host-samples
- Minimal Sample:
  - https://github.com/TheJuanitoLearnsShow/EmbedPowerFX-Session

How to run the PowerFX interpreter from .NET

- Add NuGet packages
- Initialize engine
- Add variables and formulas
  - Must have valid syntax
  - Language definition in progress:
    - https://github.com/microsoft/Power-Fx/blob/main/docs/overview.md
- Evaluate formulas

Demo

Custom Functions

- Derive from ReflectionFunction
  - Base constructor takes in:
    - Name
    - Return Type
    - Param Types
- Implement “Execute” method
- Function returning:
  - Primitive
  - Record
  - Table
- Async functions

Demo

The power of PowerFX in your apps!

PowerFX is a popular, powerful and business friendly language

Microsoft provides an interpreter for .NET

Your app can evaluate PowerFX formulas at runtime

Questions, Questions, Questions???

- Best part of sessions like this one ☺
- LinkedIN: https://www.linkedin.com/in/juantarquino
- Sample repo and slides for this session:
  - https://tinyurl.com/PowerFX-NET
- Feedback:

Embed PowerFX in .NET

Thank you! Win something at the raffle!